

Methods

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The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots, Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests and ACF/PACF plots of the two series were used to check for trends, changes in variance and overall appropriateness for ARIMA modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

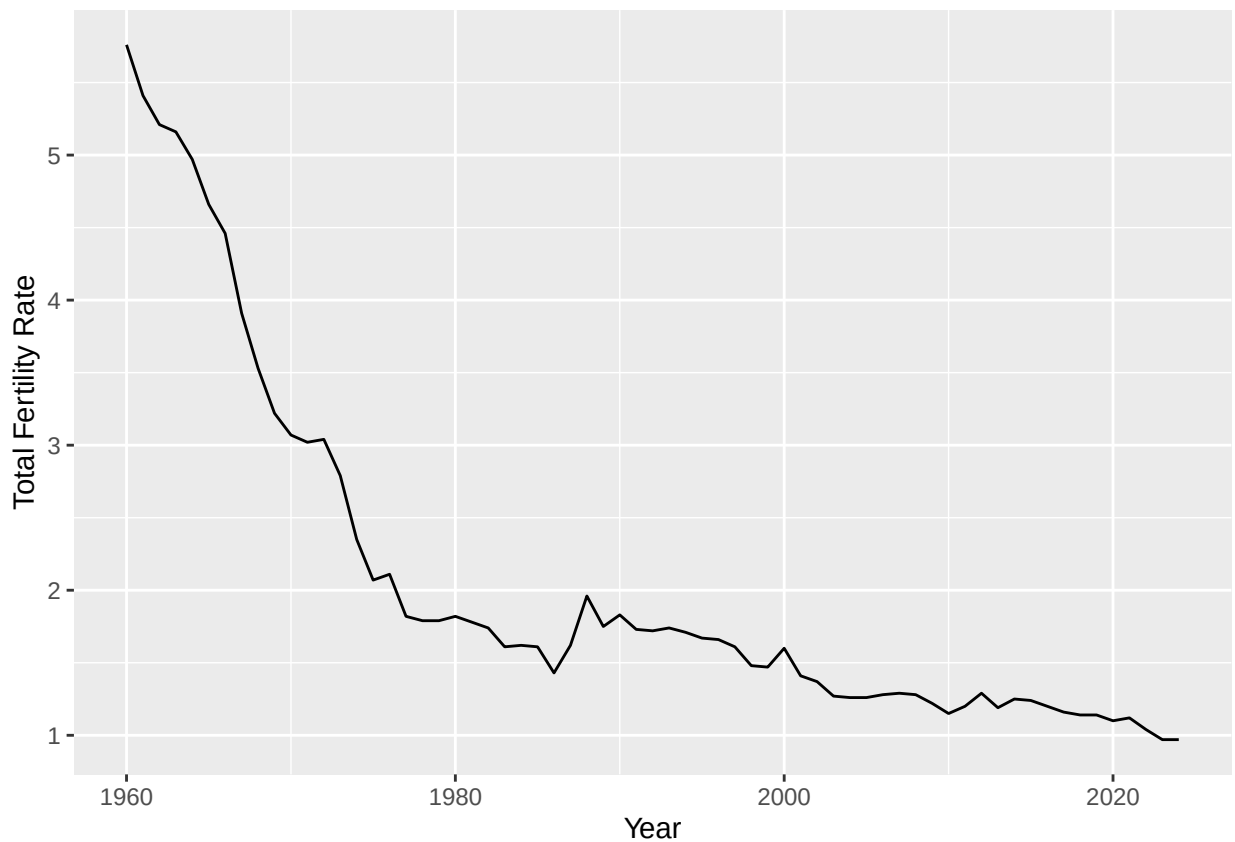


Figure 1: Annual Total Fertility Rate in Singapore

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test, residual plots and looking at the ACF and PACF plots of the series to ensure that the transformation improves the stationarity.

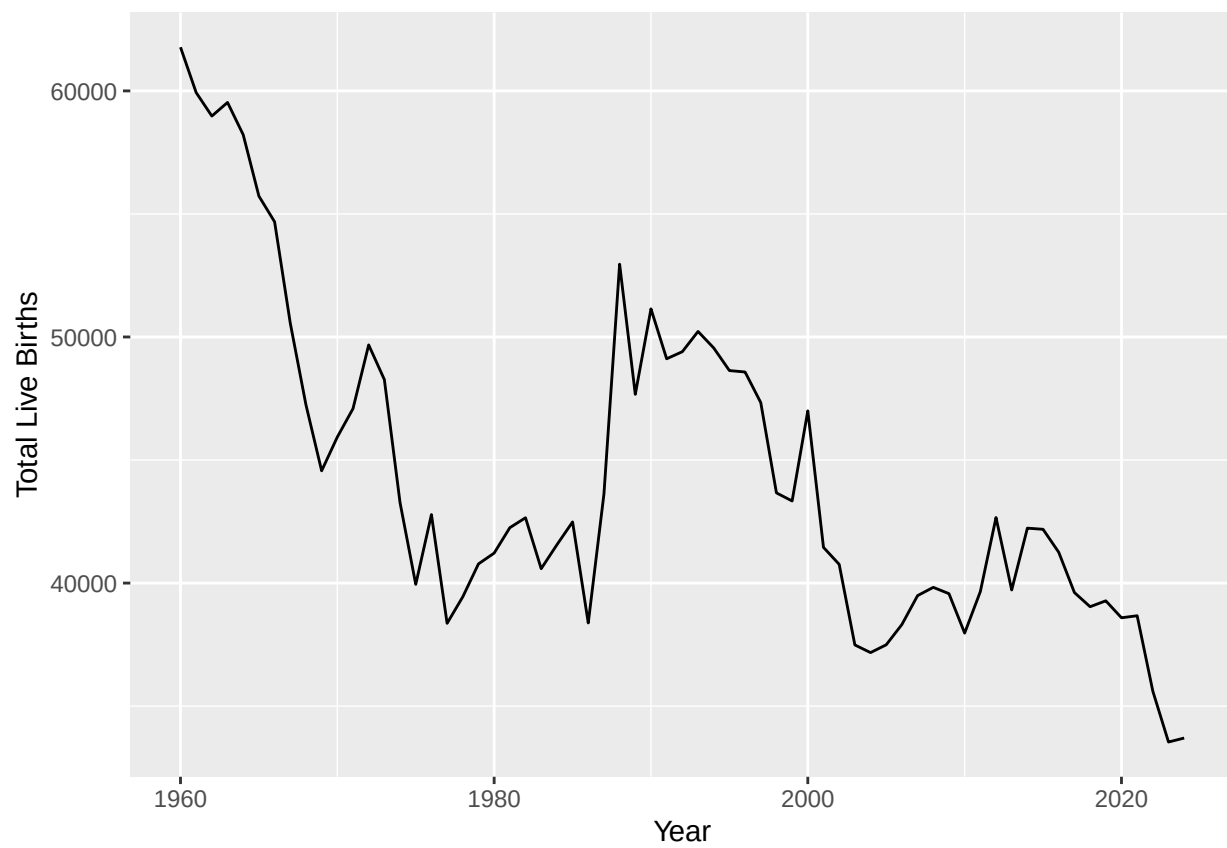


Figure 2: Total Live Births in Singapore

The KPSS test is used for testing whether a given time series is stationary (Should I go into more detail or is this the level or should it be in the appendix if it is?). The time series' are logged to scale the variance proportionally if there is a trend so to keep relative magnitude of changes in the data in the mind of the model. Differencing is used for the removal of the trends shown inside the data.

Two transformations showed potential as being stationary (first-differenced and logged, and second differenced logged). The TFR data's KPSS test gave two different p-values for the two differencing settings $p_{\log D=1} = 0.0406$ and $p_{\log D=2} = 0.1$. The logged first difference setting's p-value was below the 0.05, however, only slightly so it may still be considered if a potential model can produce white-noise residuals. For the TLB data both p-values were over the significance threshold with $p = 0.1$ for both of them meaning that we can retain the null hypothesis, such that the TLB data is stationary with either of the transformations.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set.

The next step was to pick appropriate models for forecasting for the testing period. ACF and PACF plots were used to see if the time series had hidden correlation which had carried thorough the transformations. Each time series transformation showed some dependence around lags 11-13 with the most significant spike at lag 12. This potential seasonality made us consider models which would account for this correlation. Models were only considered viable only if their residuals were approximately white noise. Among viable models, the final model selection was based on AIC and forecast accuracy over the 2013-2024 period.